

RESEARCH ARTICLE

Magnetic Resonance in Medicine

Adaptive model-based Magnetic Resonance

Inbal Beracha¹ | Amir Seginer² | Assaf Tal¹  ¹Department of Chemical and Biological Physics, Weizmann Institute of Science, Rehovot, Israel²Siemens Healthcare Ltd., Rosh Ha'ayeen, Israel

Correspondence

Assaf Tal, Department of Chemical and Biological Physics, Weizmann Institute of Science, Rehovot 7610001, Israel.

Email: assaf.tal@weizmann.ac.il

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Purpose: Conventional sequences are static in nature, fixing measurement parameters in advance in anticipation of a wide range of expected tissue parameter values. We set out to design and benchmark a new, personalized approach—termed *adaptive MR*—in which incoming subject data is used to update and fine-tune the pulse sequence parameters in real time.**Methods:** We implemented an adaptive, real-time multi-echo (MTE) experiment for estimating T_2 s. Our approach combined a Bayesian framework with model-based reconstruction. It maintained and continuously updated a prior distribution of the desired tissue parameters, including T_2 , which was used to guide the selection of sequence parameters in real time.**Results:** Computer simulations predicted accelerations between 1.7- and 3.3-fold for adaptive multi-echo sequences relative to static ones. These predictions were corroborated in phantom experiments. In healthy volunteers, our adaptive framework accelerated the measurement of T_2 for n-acetyl-aspartate by a factor of 2.5.**Conclusion:** Adaptive pulse sequences that alter their excitations in real time could provide substantial reductions in acquisition times. Given the generality of our proposed framework, our results motivate further research into other adaptive model-based approaches to MRI and MRS.

KEYWORDS

adaptive MR, Bayesian estimation, model-based reconstruction, qMRI, quantitative MR, real-time MRI

1 | INTRODUCTION

Since its inception, MR has become ubiquitous in a wide range of disciplines and has transformed many facets of biomedical research and medicine.^{1–4} In medical imaging, MRS and MRI provide objective measures of the composition, microstructure, and physiology of biological tissue, such as proton density (PD); T_1 , T_2 ; diffusion coefficient; and magnetization transfer ratio, often in a

quantitative manner. These tissue parameters provide powerful and platform-independent markers for a wide range of pathologies. At the heart of MRI and MRS lies the concept of a pulse sequence: a collection of RF and gradient pulses. Altering the sequence parameters, such as the TE, flip angle (FA), TR, TI, or other delays, controls the type and amount of contrast in the resulting signal. Sequence parameters are static: They are set in advance, in anticipation of a particular type of contrast and a range of

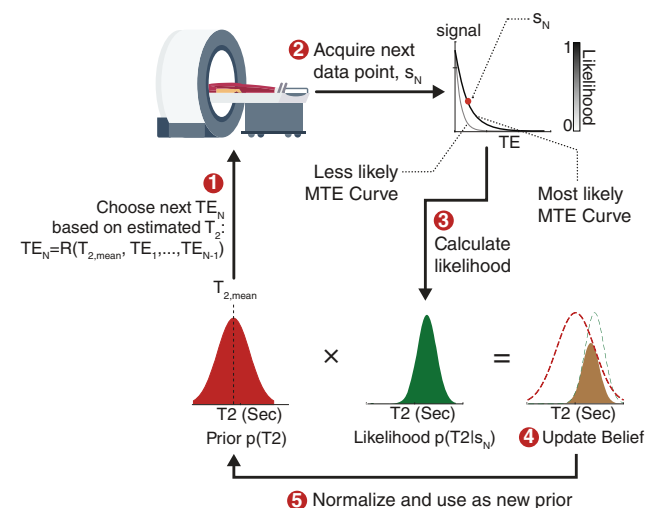


FIGURE 1 Outline of adaptive framework. The adaptive measurement scheme starts out from a prior distribution $p(T_2)$ for the estimated parameter, T_2 , indicating how likely believe we each value of T_2 is (initially $p(T_2)$ is uniform; we omit PD for simplicity). We choose our next TE using some predetermined rule $TE_N = R(T_{2,mean}, TE_1, \dots, TE_{N-1})$ based on the current probable value of T_2 , $T_{2,mean} = \int p(T_2) T_2 dT_2$ and previously measured TEs. For example, the new TE can be chosen to approximate an optimal distribution of TEs based on our estimate of T_2 . New data is acquired from the subject at the chosen TE, $s_N = s(TE_N)$ and used to calculate the likelihood $L(T_2|s_N)$ of each possible T_2 value given the measured signal s_N . The belief is updated by multiplying the likelihood by the initial prior and normalizing, and this is used as the initial prior $p(T_2)$ for the next measurement. p , probability distribution; PD, proton density; TE_N , N th TE.

biological tissue parameters (e.g., T_1 or T_2), and are varied between excitations but not between subjects.

We depart from the conventional static framework of MR and propose an adaptive approach to MR, where incoming patient data is used in real time to update the sequence excitation parameters and personalize the tissue contrast to the subject inside the MRI scanner. This approach operates by maintaining a probability distribution p that quantifies our a priori belief about obtaining different values for the tissue parameters of interest (e.g., T_2). The distribution, $p(T_2, \dots)$, which starts out as uniform, is updated after each excitation, using the incoming signal to estimate the likelihood of different values of the tissue parameter and eventually applying Bayes' rule (Figure 1). The updated distribution is used in turn to inform and sculpt the next excitation, creating a continuous, real-time interplay between the sequence and tissue parameters. The update is carried out via a calculation that is both straightforward and computationally efficient, allowing our adaptive approach to be applied to a wide range of contrast types. The proposed framework is independent of existing techniques, such as MR fingerprinting,^{5–12} parallel

imaging,^{13–16} compressed sensing,^{17–20} and model-based reconstruction,^{21,22} and in principle can be combined with each without impairing their performance.

The adaptive framework is applied to a multi-echo (MTE) experiment, and to the problem of measuring the T_2 of NAA, a small metabolite found primarily inside neurons in the central nervous system.²³ Relaxation times are markers of tissue microstructure, viscosity, and composition.^{24–29} Many metabolites are confined primarily to the intracellular space of specific cell types, for example, NAA to neurons,²³ or myo-inositol to astrocytes,³⁰ and their relaxation times specifically reflect the intracellular microenvironments of these cell types. Indeed, reports of altered metabolite relaxation times have been made in a wide range of neuropathologies, from multiple sclerosis³¹ to brain tumors^{32,33} and Alzheimer's disease.^{34,35} A recent comprehensive analysis of the existing literature of metabolite relaxation times³⁶ has shown that the relaxation times of NAA, Cr, and Cho constitute powerful potential clinical biomarkers. The analysis found that incorporating metabolite relaxation times with their concentrations would increase the area under the receiver operating characteristic curves of MR spectroscopy from 0.68 to 0.92, turning MRS from a “poor” classifier into an “outstanding” one.³⁷ However, the low sensitivity of MRS often precludes the measurement of metabolite relaxation times in practice, and, as such, clinical protocols rarely, if ever, include them.

Using computer simulations, phantom measurements, and in vivo data, we show our adaptive approach yields approximately 1.3–2.0-fold improvement in sensitivity without impairing its accuracy. Because sensitivity can be traded for acceleration, this increase is equivalent to a $(1.3)^2$ -to- $(2.0)^2$ -fold reduction in experimental time. Specifically, we demonstrate that an adaptive MTE sequence can yield the same precision for T_2 in vivo as an optimal static sequence using only 40% of the data and accelerating acquisition 2.5-fold. The principles demonstrated herein are not confined to T_2 estimates of NAA, and are universal and applicable to any model-based acquisition for both water and metabolite data.

2 | THEORY

2.1 | MTE T_2 estimation

We illustrate the proposed adaptive framework using the well-established MTE experiment for measuring T_2 , which acquires multiple TEs, one per excitation, while varying TE. The MTE signal depends on the tissue (T_2 , PD) and

sequence (TE) parameters as follows:

$$s(\text{TE}) = \text{PD} \cdot \exp\left(-\frac{\text{TE}}{T_2}\right). \quad (1)$$

Note that PD here is proportional to the proton density, but its explicit value is merely an arbitrary scaling constant. At the core of our adaptive approach is a probability distribution $p(T_2, \text{PD})$, which quantifies the assumed a priori probability of each value of the tissue parameters. Given such a distribution, we can estimate T_2 via $\langle T_2 \rangle = \int T_2 \cdot p(T_2, \text{PD}) dT_2 d\text{PD}$, which can be used, alongside all previously selected TEs, to select the N th TE (TE_N) according to some predefined rule, $\text{TE}_N = R(\langle T_2 \rangle, \text{TE}_1, \dots, \text{TE}_{N-1})$. For example, as described below, we can set R to select the TE, which minimizes the Cramer Rao Lower Bound (CRLB) for the estimated $\langle T_2 \rangle$. A new data point $s_N = s(\text{TE}_N)$ is then acquired and used to calculate the likelihood of each possible T_2 and PD value given the new measurement: $L(T_2, \text{PD}|s_N)$. If the data has Gaussian white noise with zero mean and variance σ^2 , then

$$L(T_2, \text{PD}|s_N) = \exp\left(-\left(\frac{s_N - s(\text{TE}_N)}{\sqrt{2}\sigma}\right)^2\right). \quad (2)$$

The SD of the noise, σ , can be easily and accurately estimated from an initial fast excitation with all RF pulses turned off. The distribution $p(T_2, \text{PD})$ is then updated via Bayesian estimation by multiplying the likelihood by the initial prior and normalizing, and the adaptive cycle can begin anew. This approach is illustrated in Figure 1 for the simple 1D case of a single tissue parameter, T_2 , fixing $\text{PD} = 1$.

2.2 | Choosing an update rule

A straightforward real-time update rule R chooses TE_N in a way that approximates the optimal distribution of TEs for the estimated $\langle T_2 \rangle$, in the CRLB sense. The CRLB provides a lower bound attainable by an unbiased estimator of T_2 , which is often attained in practice when using least squares fitting. For the MTE model in Equation 1, the CRLB is a 2×2 covariance matrix for the parameter of interest T_2 and the nuisance parameter PD. It is obtained by inverting the 2×2 Fisher information matrix, the elements of which are:

$$F_{ij}(T_2, \text{PD}) = -\frac{1}{\sigma^2} \sum_{k=1}^{N_{\text{TE}}} \frac{\partial s_k}{\partial \theta_i} \frac{\partial s_k}{\partial \theta_j} \quad (i, j = 1, 2), \quad (3)$$

Where $\theta_1 \equiv T_2$, $\theta_2 \equiv \text{PD}$, $s_k \equiv s(\text{TE}_k)$, and N_{TE} is the number of TE acquired, equal to the number of excitations. One

can calculate in advance the optimal distribution of TE for each value of T_2 and a fixed value of N_{TE} by minimizing the second diagonal element of the CRLB matrix, which corresponds to the variance of T_2 (see Supporting Information for details). The rule R then chooses the next adaptive TE_N such that the total distance (in, e.g., the l_1 -sense) of the distribution of measured and optimal TEs is minimal. This rule can be used in real time because it is computationally efficient and easy to implement.

Minimizing the variance of T_2 as a function of $(\text{TE}_1, \text{TE}_2, \dots, \text{TE}_N)$ for $N=20$, under the condition $\text{TE}_{\min} \leq \text{TE}_j \leq \text{TE}_{\max}$ with $\text{TE}_{\min} = 40$ ms and $\text{TE}_{\max} = 800$ ms, yields a bimodal distribution of TEs, placed at a 1:5 ratio between $\text{TE}_{\min}$ and approximately $1.4 \times T_2$. This solution holds to an excellent approximation for all physiological values of T_2 between 0.05 and 0.45 s, and was encoded into the update rule R of the adaptive sequence. The first acquisition, however, was always set to acquire the same TE: $\text{TE} = \text{TE}_{\min} = 40$ ms, after which the adaptive framework and update rules guided subsequent selection of TEs.

2.3 | Comparing adaptive and static approaches

A fair comparison between static and adaptive approaches should be carried out by fixing the total measurement time and comparing the precision (SD) and bias (estimated minus true) for T_2 . We optimized the static sequences to minimize the mean CRLB over a range of physiologically relevant values of T_2 . The CRLB has been used extensively as a cost function in designing pulse sequences for magnetization transfer,³⁸ MR fingerprinting,³⁹ k-space sampling,⁴⁰ thermometry⁴¹ and even specifically for the problem of selecting optimal TEs for mapping T_2 .⁴²

To ensure fairness, the optimization must assume the same initial prior belief as the adaptive approach, which in our case is a uniformly distributed between $T_2 \in [0.05, 0.45]$ s. This range was derived from the reported range of literature values for NAA T_2 : Wyss et al. report⁴³ a T_2 range for the NAA singlet at 2.01 ppm of approximately 100–400 ms in healthy volunteers. NAA T_2 s as high as almost 500 ms have also been reported in the white matter of young adolescents.⁴⁴ This is in line with a recent review that surveyed metabolite relaxation times at 1.5 T and 3.0 T across a wide range of pathologies and regions of interest.³⁶

We discretized T_2 in M equispaced steps between its minimum (0.05 s) and maximum (0.45 s) values. We calculated the average CRLB matrix over all values of T_2 and minimized its second diagonal element divided by the value of T_2 — corresponding to the mean squared

coefficient of variation of T_2 (i.e., squared relative error in T_2)—to obtain a set of optimal TEs for the static case:

$$\min_{TE_1, \dots, TE_{N_{TE}}} \frac{1}{M} \sum_{k=1}^M \frac{\text{Var}(T_{2,k})}{T_{2,k}^2} \quad \text{s.t. } TE_{\min} \leq TE_k \leq TE_{\max} \quad (k = 1, \dots, N_{TE}). \quad (4)$$

The optimization was carried out for $N_{TE} = 20$. TR was fixed at 6 s for both the static and adaptive approaches to allow the spins sufficient time to return to equilibrium, resulting in a total acquisition time of 2 min. Additional parameters were $TE_{\min} = 0.04$ s, $TE_{\max} = 0.8$ s. The optimization problem was carried out 20 times using an interior point-constrained algorithm in MatLab 2021b (MathWorks, Natick, MA) (the `fmincon` function), starting each time from a different initial random guess. The best overall solution was taken, and only if a majority of solutions agreed (if not, the optimization was rerun). This ensured our TEs represented global maxima. This yielded a trimodal distribution of TEs, which was sharply peaked at around 0.04, 0.11, and 0.37 s, at a TE sampling ratio of 1:8:11.

3 | METHODS

3.1 | Numerical simulations

Monte-Carlo simulations were used to compare the bias and SD of static and adaptive approaches for the MTE experiment. Additional simulations are provided in the Supporting Information for another experiment type, an inversion recovery (IR), which is designed to estimate the T_1 (Figure S1). The temporal evolution of the prior $p(\text{PD}, T_2)$ throughout a sample simulated adaptive acquisition is also showcased in the Supporting Information for $T_2 = 200$ ms, SNR = 20 (Figure S5).

Both the static and adaptive approach fixed the total acquisition time at 2 min, and TR at 6 s. This allowed each approach to sample 20 TEs. The SNR per excitation was fixed at 20 at $TE = 0$ ms. The static MTE experiment used the optimal static TEs, placed at 40, 110, and 370 ms at a 1:8:11 ratio (obtained by optimizing Equation 4). Both simulations—adaptive and static—fixed T_2 and ran a total of 2000 experiments, each with different normally distributed generated noise. Each experiment yielded a single estimate for T_2 ; the distribution of 2000 T_2 s was used to calculate the SD and bias of T_2 for each approach. This was repeated for different fixed values of T_2 between 0.05 and 0.45 s.

The adaptive acquisition was used to guide the real-time selection of TEs. However, to avoid biases related to the parameter estimation procedures, both the adaptive

and static data were fit to the decaying exponential model (Equation 1) using a straightforward least-squares approach.

Real-time computational costs might preclude complex fitting or modeling procedures to be undertaken while the sequence is running; simplified models might be used instead, and the acquired dataset can be reprocessed offline once the acquisition is overusing a more complete and accurate model. To investigate the effects of possible estimation biases in T_2 incurred in real time might have on the final T_2 offline processing, we ran Monte Carlo simulations using an ad hoc T_2 bias during real-time estimation. The true value of T_2 in this simulation was assumed to be $T_2 = 0.2$ s.

3.2 | Phantom experiments

Precision and bias of static and adaptive T_2 estimations were validated on a spherical phantom, containing 1.9 L water doped with Gadolinium-DTPA (0.5 mM) and metabolite concentrations similar to in vivo human brain: 12.5 mM NAA, 2 mM choline chloride, 10 mM creatine anhydrous, 7.5 mM myo-inositol, 10 mM L-glutamic acid, 5 mM sodium lactate, and 2 mM γ -aminobutyric acid. Sodium azide (NaN₃ 0.01%) was added to avoid microbial growth. pH was adjusted by means of sodium phosphate buffer to be 7.2. For the phantom, prolonged 3.5-h “gold-standard” MTE measurements yielded $T_2 = 660$, 545, and 347 ms for the NAA, Cr, and Cho singlets at 2.01, 3.03, and 3.19 ppm, respectively. Because the T_2 s of NAA and Cr were longer than the assumed prior maximum of 450 ms, the adaptive scheme targeted the Cho peak (as opposed to the in vivo measurements, which targeted the NAA peak; see below).

The SD of T_2 was computed for the static and adaptive approaches. The ratio of the static to the adaptive calculation was formed, and the 95% confidence interval (CI) for ratio was calculated. A two-tailed F -test was used to compare the variances of the two methods against the null hypothesis that they are equal.

For a phantom, given m adaptive and n static measurements, the 95% CI ($\alpha = 0.05$) on the ratio of two SDs is:

$$CI = \left[\sqrt{\frac{1}{F_{\frac{\alpha}{2}}(n-1, m-1)}}, \sqrt{\frac{1}{F_{\frac{\alpha}{2}}(m-1, n-1)}} \right] \cdot \frac{\sigma_{\text{static}}}{\sigma_{\text{adaptive}}},$$

where F is the F -statistic. Setting $n = m = 60$ and assuming a ratio $\sigma_{\text{static}}/\sigma_{\text{adapt}} = 1.6$ for Cho (based on simulations, using $T_{2,\text{Cho}} = 347$ ms in the phantom), the CI was [1.24, 2.07], that is, sufficiently narrow to determine whether the two approaches differed in their precision.

All data was acquired on a Siemens 3 T Prisma scanner (Siemens Healthineers, Erlangen, Germany), using the manufacturer-provided 20-channel head coil for reception and the body coil for transmission, capable of maximal RF amplitude of 19 μ T. We have used an in-house implementation of a point-resolved-spectroscopy localization sequence. The adaptive protocol was implemented by using a real-time feedback mechanism built into the image reconstruction framework of Siemens (Siemens Healthineers) (Image Calculation Environment [ICE]), conventionally used for real-time navigators and on-the-fly image reconstruction. This allowed us to store and update the prior on T_2 , $p(T_2, PD)$, change TE in real time using ICE code written in C++, and store real-time TEs and T_2 estimations in the Siemens (Siemens Healthineers) scanner real-time log files for offline processing. The bandwidths of the excitation and refocusing pulses were 3.6 and 1.1 kHz, respectively, and their maximal amplitudes were both set at $\gamma B_1 = 0.8$ kHz. Water suppression took 426 ms and was achieved by applying a set of interleaved water-selective pulses and crusher gradients prior to excitation, with variable flip angles and delays optimized in-house.

An additional control static dataset was also acquired from the phantom, choosing TEs based on our a priori knowledge of the true value of $T_2 = 347$ ms at a 1:5 ratio between $TE_{\min} = 40$ ms and $1.4 \times T_2 \approx 485$ ms. As with the numerical simulations, this represents an unrealistic scenario, and allows us to estimate the best possible performance one could ever expect for estimating T_2 (in a CRLB sense). As with the static and adaptive cases, 60 repetitions were acquired.

Even though the adaptive scheme employs real-time Bayesian statistics to inform its choice of parameters, once acquired, its data can be processed using an identical pipeline to that of the static acquisition. This has been done for our simulations, and phantom and in vivo datasets, to avoid differences relating to the processing pipeline. All processing used MatLab 2021b (MathWorks). A singular value decomposition was used to combine the coils. Datasets were zero-filled fourfold along the spectral domain and apodized with a Gaussian window with a FWHM of 5 Hz. Phasing was achieved by summing all averages and maximizing the area underneath the real part of the spectrum between 0.5 and 4.2 ppm. Signal amplitudes used the maximum of the real part of the peak. Fitting employed a simple least-squares routine as implemented by the MatLab 2021b (MathWorks) `fminsearch` function.

3.3 | In vivo experiments

We validated the predicted gains of our proposed adaptive approach in a cohort of seven healthy volunteers

who provided informed consent, as approved by Wolfson Medical Center Helsinki committee. Each volunteer was initially positioned head-first supine. Following an anatomical T_1 -weighted MPRAGE scan, a $1.5 \times 1.5 \times 1.5$ cm³ voxel was placed in parietal white matter and shimmed using in-house routines. Twelve adaptive and twelve static acquisitions were interleaved, each lasting 12 min ($12 \times 2 = 24$ min per approach). Total scan time was under 60 min. The adaptive scans targeted the T_2 relaxation time of the NAA singlet at 2.01 ppm. As for the phantom experiments, the adaptive acquisitions were all prospective and updated TE and the prior $p(PD, T_2)$ in real time.

Both the adaptive and static acquisitions were processed offline as detailed for the phantom experiments. Each acquisition yielded a single T_2 estimate for NAA, resulting in twelve T_2 estimates per approach (static, adaptive) and per volunteer. A linear mixed model was used to test for statistically significant differences between the two approaches, as well as estimate the intrasubject SD for each approach, using the model $T_2 \sim 1 + \text{method} + (1|\text{subj})$. The linear mixed model was used to obtain the intrasubject SDs, as well as their 95% CIs.

A Monte-Carlo simulation of our mixed-model approach was used to determine the number of volunteers required, assuming $\sigma_{\text{static}}/\sigma_{\text{adaptive}} = 1.5$, which is expected³⁶ for the typical in vivo white matter NAA T_2 s of around 350 ms at 3 T. This yielded a CI of $\sigma_{\text{static}}/\sigma_{\text{adapt}} \in [1.24, 2.34]$.

The superior precision per unit time of adaptive sequences can be traded for accelerated data acquisition for a given desired precision by simply acquiring fewer excitations. To determine the commensurate acceleration offered by the adaptive approach, we recalculated the mean precision σ_{adapt} of the adaptive MR dataset by taking successively fewer excitations (omitting later excitations), until further removal of excitations resulted in worse precision by the adaptive approach compared to the static approach ($\sigma_{\text{adapt}} > \sigma_{\text{static}}$). The excitations were always omitted starting from the last and working backward. That is, only the initial adaptive TEs were used, as would be done in an actual experiment.

4 | RESULTS

4.1 | Numerical simulations

Simulation results for MTE experiments are shown in Figure 2. Figure 2A shows the optimal static TEs that minimize the average T_2 CRLB (Equation 4) over the range $T_2 \in [0.05, 0.45]$ s. Figure 2B shows the performance of the adaptive method for 2000 iterations of the adaptive

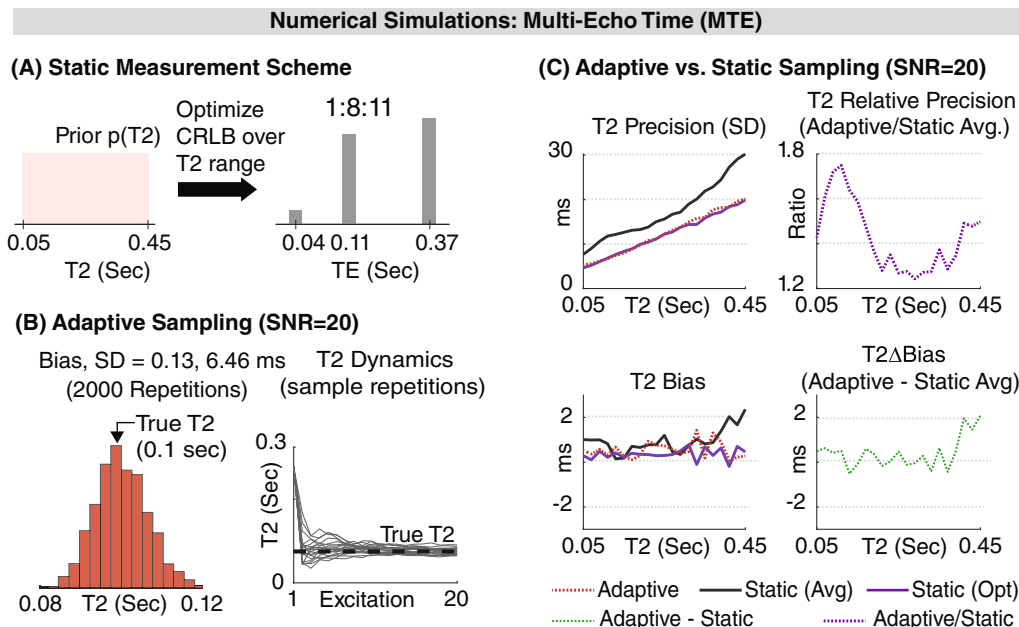


FIGURE 2 Numerical simulations. (A) The optimal TEs that minimize the mean coefficient of variation of T_2 over a range of prior values, $T_2 \in [0.05, 0.45]$ s, for a 120 s static acquisition. (B) The performance of the adaptive scheme for a specific value $T_2 = 0.1$ s, for 2000 repeated iterations, each with different Gaussian noise, assuming SNR = 20 for a single excitation at TE = 0 s. Shown is the histogram of estimated T_2 values from the 2000 iterations; its width is the SD of T_2 , whereas the difference between its center and the true T_2 is the bias. This histogram shows the adaptive estimation is unbiased. Also shown are the temporal dynamics of T_2 for a subset of 20 sample runs, showing that T_2 converges to around its true value (dashed black line) very early on in the scan. (C) Comparison of the precision and bias of three experiments using Monte Carlo simulations: The solid black line corresponds to the optimal static experiment, with TEs chosen as in (A). The dotted red line corresponds to the adaptive scheme. Finally, the solid purple line represents a static acquisition, with TEs optimized for each specific single value of T_2 in a Cramer Rao sense; this case is obviously unrealistic because one does not know in advance the value of the T_2 but provides the best possible precision and bias one could hope to attain. These plots show the performance of the adaptive scheme (dotted red) is almost identical to that of the optimal (solid purple) approach. This is a result of the adaptive scheme converging very early on close to the true parameter values (as shown in (B)), resulting in near-optimal selection of TEs

approach, each taking 120 s and using different randomly generated Gaussian noise, for $T_2 = 0.1$ s. The histogram shows the distribution of the 2000 estimated T_2 s, indicating the approach is unbiased. The dynamics of T_2 are also shown for a subset of the iterations, showing they converge very rapidly to the vicinity of the true value. Figure 2C shows both approaches—static and adaptive—are unbiased, whereas the adaptive approach offers a 1.25–1.7-fold increase in precision for T_2 , equivalent to a commensurate 1.6–2.9-fold acceleration. Figure 2C also shows the performance of the adaptive scheme (dotted red line) is almost identical to the best possible precision attainable, had we known in advance the true value of T_2 and optimized the TEs for that specific value in the CRLB sense (i.e., chosen TEs to minimize the CRLB calculated for a single, specific value of T_2). Such an optimal experiment is obviously unrealistic because one does not know a priori the true value of the estimated tissue parameter before running the experiment but serves as a benchmark for the best possible outcome one might hope for. Additional simulations for an IR experiment are shown in the Appendix and support

an even larger gain in precision and acceleration for T_1 estimation.

Our computer simulations also confirmed the adaptive T_2 approach is robust even in the face of substantial real-time errors. Figure 3 shows that even a 20% (0.04 s) real-time bias in T_2 (true value: 0.2 s) leads to a negligible $\leq 3\%$ reduction in its precision (Figure 3A) and no discernible effect on its bias (Figure 3B) when correctly estimated off-line using the measured data.

4.2 | Phantom experiments

The phantom experiments confirm the gains to precision predicted by the numerical simulations. Figure 4 shows experimental results from the phantom experiments, consisting of 60 repetitions of both the static and adaptive 2-min acquisitions. The adaptive acquisitions targeted the choline singlet at 3.2 ppm ($T_{2,\text{Cho}} = 347$ ms). Voxel placement is shown in Figure 4A, and a sample single excitation is plotted in Figure 4B, showing typical spectral

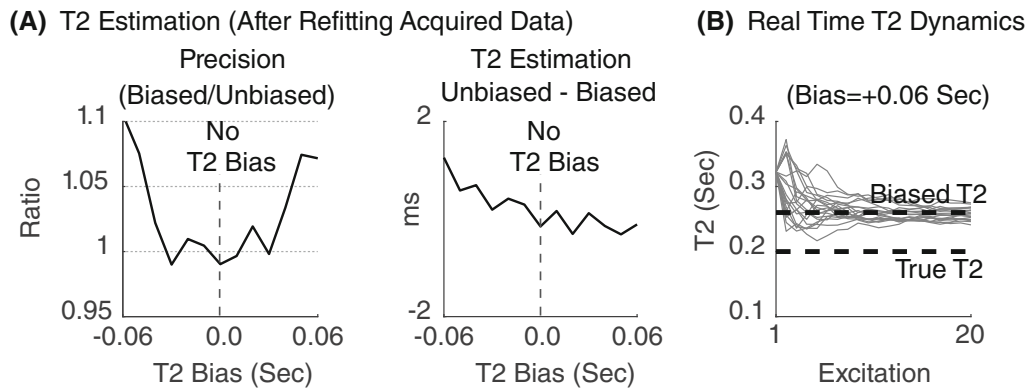


FIGURE 3 The effect of T_2 real-time bias. Numerical simulations (assuming SNR = 20 as in Figure 2), demonstrating that even significant real-time biases in T_2 lead to a minimal loss of precision. (A) Precision and bias of the T_2 , estimated off-line, as a function of the bias in estimating T_2 in real time, under the assumption that the signal is correctly modeled off-line using more advanced and computationally intensive model (shown here is the case where $T_2 = 0.2$ s). Even large real-time biases of 20% (i.e., 0.04 s) result in almost no loss of precision or bias in the offline processing. (B) The real-time dynamics of T_2 for a bias of 30% (+0.06 s), showing the real-time acquisition, converges to the biased value of T_2 early on

quality and SNR. The acquired data from a single static and adaptive acquisition is shown in Figure 4C,D, with the fitted MTE curve (Equation 1). Note how the static data points are acquired at the optimal TEs shown in Figure 2A, whereas the adaptive data points are distributed unequally between the minimal TE ($TE_{\min} = 40$ ms) and $TE \approx 1.4 \times T_2$. This distribution approximates the optimal one for specific value of $T_{2,\text{Cho}} = 347$ ms of Cho. Both the adaptive (red) and static (gray) acquisitions provide unbiased estimates of the true T_2 (Figure 4F), but the adaptive acquisitions yield better 1.76-fold better precision: $\sigma_{\text{adapt}} = 50$ ms vs. $\sigma_{\text{static}} = 88$ ms, with corresponding CIs $\sigma_{\text{static}}/\sigma_{\text{adapt}} \in [1.24, 2.07]$. An F -test shows the ratio of these SDs to differ from unity in a statistically significant manner. Figure 4F also shows that the adaptive performance (red) is equivalent to the best possible performance one could ever expect from a CRLB-based estimator (magenta), for the unrealistic scenario in which the TEs are chosen to minimize the CRLB using a priori knowledge of the single true value of T_2 .

4.3 | In vivo experiments

Figure 5A,E showcases sample data from a single volunteer, including voxel placement, a typical single excitation at $TE = 40$ ms (following a 4 Hz apodization, also used in real time), fitted MTE curves for a single repetition of both the adaptive and static sequences, and the dynamics of T_2 estimation for all 12 adaptive repetitions. Figure 5F shows the distribution of fitted T_2 values of NAA for each of the seven volunteers, providing visual indication the means are not significantly different in the statistical sense, and that the SD of the adaptive scheme is overall better than

that of the static one (i.e., smaller error bars). The ratio $\rho = \sigma_{\text{static}}/\sigma_{\text{adapt}}$ of SDs for each volunteer is shown above each bar plot. The mean ratio of SDs, averaged over all volunteers, is $\sigma_{\text{static}}/\sigma_{\text{adapt}} = 1.68$, that is, approximately 70% better for the adaptive approach. A mixed model analysis confirmed no statistically significant differences between the means of the two groups and provided the following means and 95% CIs for the SD of T_2 for each approach: $\sigma_{\text{static}} = 49.5$ ms, $CI_{\text{static}} = [41.6, 59.0]$ ms, $\sigma_{\text{adapt}} = 29.0$ ms, $CI_{\text{adapt}} = [24.3, 34.5]$ ms. The 95% CI for the ratio of the two SDs was $[1.21, 2.42]$, confirming the adaptive approach does indeed offer a statistically significant improvement in precision over the static one.

To determine the acceleration offered by the adaptive approach, later excitations were successively removed from the dataset until its average precision (over all subjects) matched that of the static acquisition. Only eight adaptive excitations, that is, 40% of the data, were required to attain the same precision obtained by the static approach, which used 20 excitations, resulting in an acceleration of the measurement by a factor of $20/8 = 2.5$.

5 | DISCUSSION

The problem of adapting a process to incoming data in real time has been explored extensively within the field of signal processing; its best-known implementation is the Kalman filter, which updates the state of a system based on its dynamical model and sensor data.⁴⁵ Recently, several suggestions have been made for adapting the k-space sampling patterns to match specific features of the object—or class of objects—being imaged. This general problem of designing optimal pairs of samplers and reconstruction

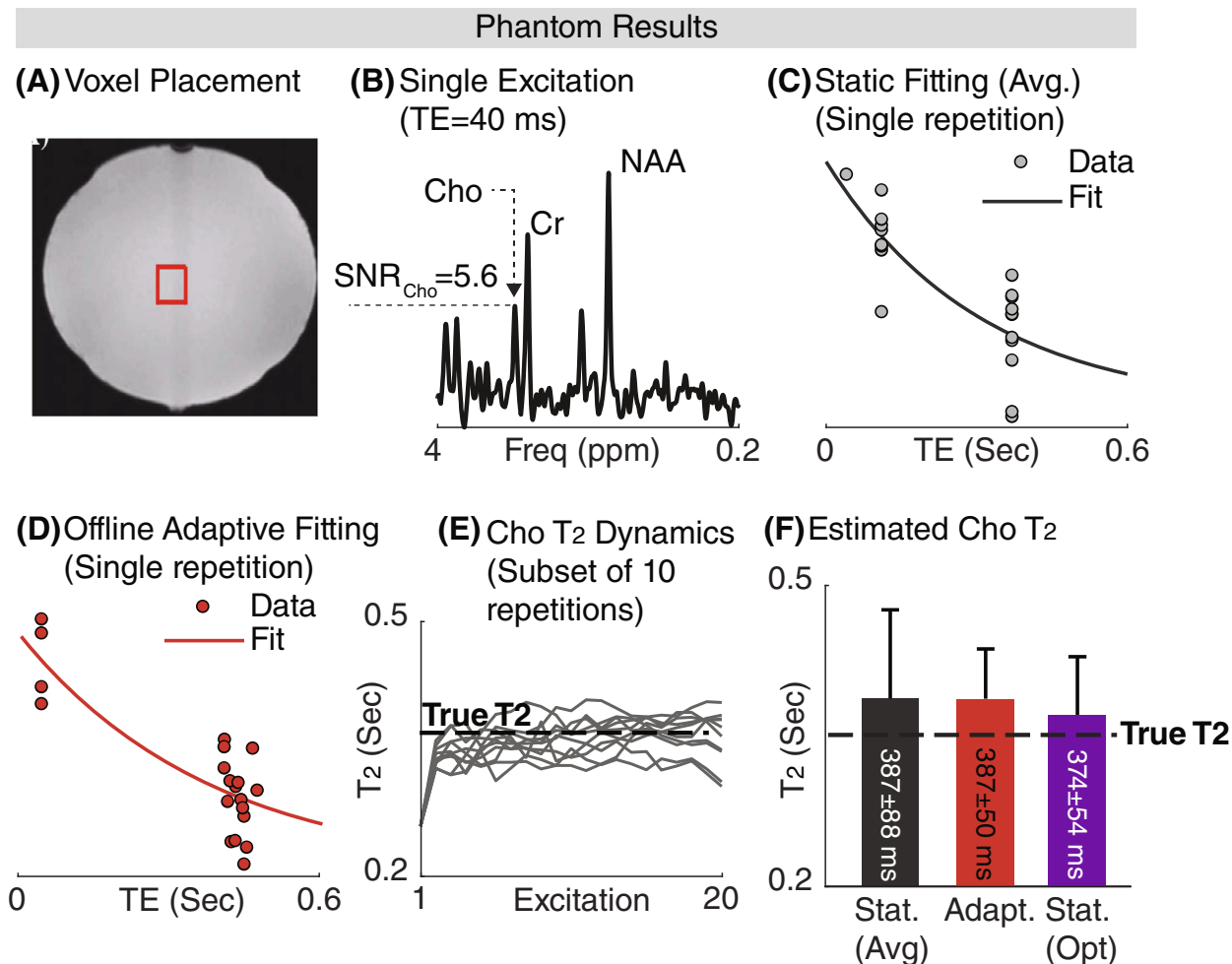


FIGURE 4 Phantom data. (A) $1.5 \times 1.5 \times 1.5 \text{ cm}^3$ voxel placed at the center of the phantom. (B) Sample spectra: single excitation with water suppression, shown for a single excitation ($\text{TR}/\text{TE} = 6000/40 \text{ ms}$). The SNR of the choline peak is 5.6. (C, D) Sample data and fitted MTE curve from a single repetition. Each acquisition consists of 20 data points, acquired over a 2-min interval. The static data has been sampled at the optimal TEs (shown in Figure 2A) $\text{TE} = 40, 110, 370 \text{ ms}$, at a ratio of 1:8:11. The adaptive data has its TEs distributed around $\text{TE} = 40, 520 \text{ ms}$ at an approximate 1:4 ratio, which is the optimal distribution of TEs (in a CRLB sense) corresponding to the true $T_{2,\text{Cho}} = 347 \text{ ms}$. (E) The dynamics of T_2 for a subset of 10 repetitions out of the 60 in total. (F) Bars plots of T_2 estimations (mean $\pm$ SD over all 60 results) for the static (gray) and adaptive (red) acquisitions. Both methods are unbiased, and the adaptive method exhibits better precision by a factor of $\sigma_{\text{static}}/\sigma_{\text{adapt}} = 88/50 = 1.76$, in line with the simulation predictions for $T_{2,\text{Cho}} = 347 \text{ ms}$. The rightmost bar (magenta) represents the result of our unrealistic control experiments, where TEs were chosen in advance using our a priori knowledge of T_2 (this is equivalent to the static/optimal case in Figure 2C). This confirms that the adaptive approach is equivalent to the best possible performance one could expect out of a CRLB estimator, when the estimated parameter is known in advance. CRLB, Cramer Rao Lower Bound; MTE, multi-echo.

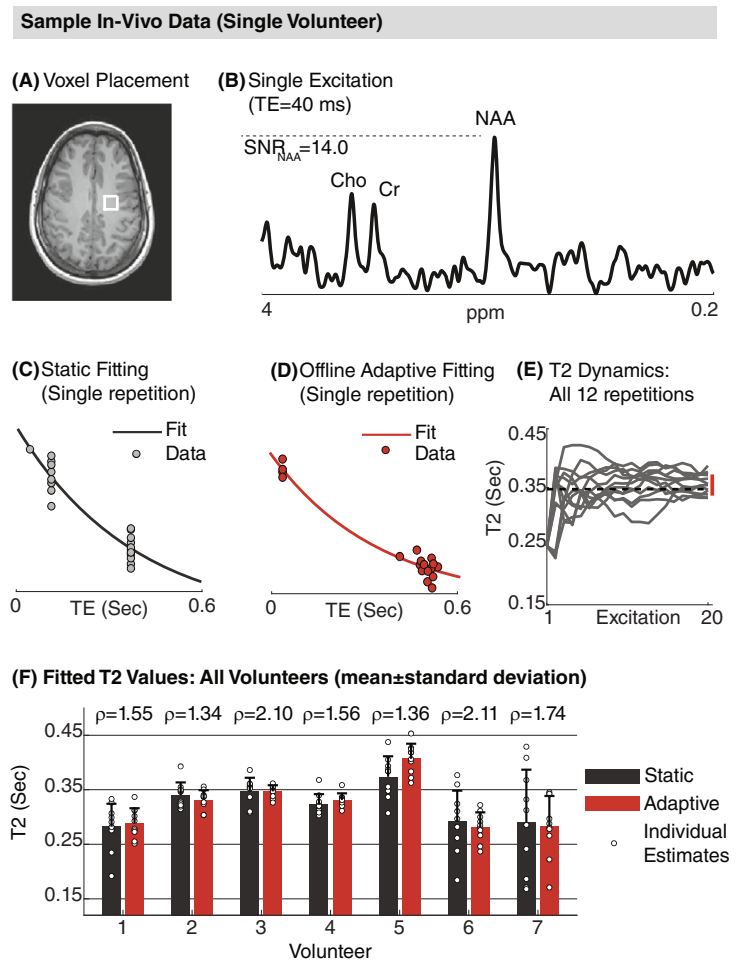
strategies has led to several adaptive k-space sampling paradigms based on machine learning, sometimes termed *active MRI* or *sequential MRI*.^{46–52} These recent approaches are therefore independent of, and complementary to, the currently proposed framework, which focuses on the excitation itself and not the Fourier sampling. In fact, data acquisition in this work was confined to a single MRS voxel and did not involve sampling of k-space at all.

By nature, such adaptive measurements are real time, but the converse is not true: real-time MRI,^{53–55} another subfield that rapidly acquires image slices at high temporal resolutions of 10 to 50 ms, contains no adaptive

component to it at all. Real-time estimation and prospective correction of subject motion is another major field of research in MR imaging, which uses navigator data or external field sensors.^{56–60} Whereas these approaches share similarity in being real time and adaptive, they do not make use of the details of the pulse sequence used to acquire the image contrast, nor alter the sequence to change the contrast itself.

Bayesian statistics provide an easy, straightforward framework for such updates. Indeed, adaptive Bayesian-based estimations have been used for measurements involving nitrogen vacancy centers, where they

FIGURE 5 In vivo data. (A) Sample voxel placement ($1.5 \times 1.5 \times 1.5 \text{ cm}^3$) in parietal WM. (B) Sample single excitation (TR/TE = 6000/40 ms), showing $\text{SNR}_{\text{NAA}} = 8.1$. (C, D) Fitted MTE curves for two sample 2-min repetitions from the sample volunteer 2-min acquisitions, showing how the TE converges early on to the correct value. (E) Distribution of T_2 estimations from both the static and adaptive approaches, showing the adaptive approach exhibits a statistically significant ≈ 1.74 -fold improvement in its SD. (F) Bar plots showing the mean and SD of T_2 for each method for each volunteer (circles represent estimations from each of the 12 repetitions). The number ρ above each pair of bar plots is the ratio of the static and adaptive SDs of T_2 for each subject. Measurements are unbiased relative to each other, but the precision of the adaptive approach is approximately 70% better than the static one. WM, white matter.



have been used to improve the precision of quantum sensors in estimating resonant spin frequencies in the presence of noise.^{61–64} Adaptive schemes are also completely independent from other sources of acceleration, such as compressed sensing, MR fingerprinting, and parallel imaging, and can therefore be combined with them without detracting from their performance. Finally, unlike static sequences, adaptive acquisitions make almost no a priori assumptions on the range of the parameter being estimated; this is unlike static sequences, which must optimize their sequence parameters (e.g., TE) in anticipation of a specific range of tissue parameter (e.g., T_2). The wider this range, the more advantageous adaptive approaches become (and more capable of handling tissue parameters outside this range; see Supporting Information, Figure S2).

5.1 | Adaptive approaches and sequence acceleration

Adaptive MR improves tissue parameter estimation in two ways: first, by informing the selection of optimal sequence

parameters (e.g., TR, TE, TI, flip angle) while the sequence is being executed, given our changing a priori knowledge about the tissue parameters (e.g., T_1 , T_2 , proton density); second, in some paradigms, by updating our prior knowledge of the tissue parameters to allow us to reduce total measurement time, or, equivalently, increase the number of measurements for a fixed total measurement time. For example, in IR experiments, the post-excitation delay needed for spins to return to thermal equilibrium is often chosen to be five times as long as the longest projected T_1 in the sample. However, if the true T_1 is significantly shorter than the longest possible T_1 , as is often the case, the post-excitation delay can be shortened dramatically and, consequently, more averages can be acquired for a fixed total acquisition time.

The improved sensitivity offered by adaptive schemes, along with the fast convergence of tissue parameters to their true values, mean that adaptive sequences can often be completed faster than their static counterparts for a given desired precision, simply by using fewer excitations. Our adaptive in vivo data produces the same precision for T_2 as the static data with only 40% of the excitations, corresponding to a 2.5-fold acceleration; and our computer

simulations for MTE and IR experiments show that acceleration of up to fourfold are possible. Because adaptive schemes maintain a distribution of the tissue parameters, for example, for T_2 in MTE, this can be used as a stopping criterion in real time once the precision of tissue parameter of interest reaches a certain level. The distribution can also be used to identify cases in which convergence is poor, for example, due to patient motion or some other artifact, and potentially prolong the acquisition and disregard poor quality data points.

5.2 | Real-time computational considerations

The use of Bayesian estimation provides an efficient framework for incorporating prior knowledge into parameter estimation in real time. In the case of a signal with a simple Gaussian or other well-modeled noise distribution, it only involves the calculation of a likelihood function and the multiplication of two probability distributions. This might become somewhat time-consuming if the signal model contains multiple parameters, in which case these probability distributions become multidimensional and might require a substantial amount of memory and central processing unit time to store and multiply. To overcome this, such a distribution can be modeled using a low dimensional representation, for example, sum of Gaussians, which can greatly enhance computational efficiency. Even using our brute-force 2D representation, we have timed the adaptive update stage to require approximately 35 ms. In cases in which such calculations are too time-consuming to carry out within a single TR, the sequence can be partitioned into acquisition windows consisting of several TRs, and each window can be processed in bulk, allowing for more time to carry out computations. Alternatively, extremely fast neural-network solvers now exist that can extract tissue parameters from the acquired signal within milliseconds, making it possible to consider adaptive schemes even for complex problems such as MR fingerprinting or multi-pool magnetization transfer models.^{21,65–71} Neural networks can be used even in cases in which a closed-form explicit model for the signal (e.g., Equation 1) does not exist.

5.3 | Limitations

Any adaptive scheme must optimize some target function. In our in vivo data, we updated the measurement parameter, TE, to minimize the variance in T_2 for a single peak, NAA. Our focus herein was specifically on a single voxel spectroscopic acquisition; in more complex multivoxel and/or imaging sequences, one must carefully

consider what to optimize. The optimization target must be tailored to each case. For example, in a BOLD-fMRI experiment, the TE could be adjusted to maximize BOLD contrast between rest and activation based on each individual's hemodynamic response. When imaging a tumor, contrast between the tumor margins and surrounding tissue on a T_1 -weighted image might be crucial, and therefore the contrast-to-noise ratio between two regions of interest in the image might be appropriate. These regions could be defined in advance based on a quick scout image, or even identified in real time using fast-performing neural networks during acquisition as part of the update process. Other optimization targets might modify the sampling in k-space based on whether the signal exhibits sufficient sparsity in some representation. The concept of adaptive sequences need not be confined to a single contrast type. When a pathology exhibits change in several contrast mechanisms, one can start out using a sequence capable of measuring more than one contrast type, such as a MR fingerprinting acquisition with a highly variable schedule, and evaluate in real time that contrast type yields the greatest contrast-to-noise ratio, or deviation from some average healthy brain template. The sequence can then be adjusted not just to maximize contrast but also to pinpoint and focus on the best type of contrast.

6 | CONCLUSIONS

Adaptive sequences, in which sequence parameters are adjusted in real time, provide a completely new avenue for accelerating MR data acquisition, improving sensitivity, and increasing image contrast. We have presented a framework for adaptive acquisitions which makes use of a signal model, Bayesian estimation, and Cramer-Rao bounds to adaptively adjust any sequence parameter in response to incoming measurements, while the subject is inside the MRI scanner. This framework presents a radical departure from the static approach to pulse sequence design, in which parameters are fixed in advance, regardless of the subject. Our results show this framework provides substantial gains to sensitivity and/or acceleration for measurements of T_2 , as well as T_1 (using simulations in the Supplementary Materials). These results motivate further research into adaptive MR for both imaging and spectroscopy.

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CONFLICT OF INTEREST STATEMENT

Dr. Amir Seginer has previously been employed by Siemens Healthineers, Israel.

ORCID

Assaf Tal  <https://orcid.org/0000-0001-6718-6522>

TWITTER

Assaf Tal  @AssafMRILab

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SUPPORTING INFORMATION

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FIGURE S1. Numerical simulations: Inversion Recovery (IR). Comparison of static and adaptive acquisitions for an inversion recovery experiment, showing expected gains between 20% and 100% to precision (i.e. accelerations between 1.4 and four-fold). (A) The optimal TIs which minimize the mean coefficient of variation of T_1 over a range of prior values, $T_1 \in [0.6, 1.6]$ Sec for a 120-s acquisition. (B) The performance of the adaptive scheme for a specific $T_1 = 0.8$ s, for 2000 repeated iterations, each with different Gaussian noise, assuming $\text{SNR} = 20$ for a single excitation. Shown is the histogram of estimated T_1 values from the 2000 iterations. This histogram shows the adaptive estimation is unbiased. Also shown are the temporal dynamics of T_1 for a subset of 20 sample runs, showing that T_1 converges to around its true value (dashed black line) very early on in the scan. (C) Comparison of the precision and bias of three experiments, using Monte Carlo simulations: The solid black line corresponds to the optimal static experiment, with TIs chosen as in (A). The dotted red line corresponds to the adaptive scheme. Finally, the solid purple line represents a static acquisition, with TIs optimized for each specific single value of T_1 ; this case is obviously unrealistic, since one does not know in advance the value of the parameter being estimated, but provides the best possible precision and bias one could hope to attain. These plots show the performance of the adaptive scheme (dotted red) is almost identical to that of the optimal (solid purple) approach. This is a result of the adaptive scheme converging very early on close to the true parameter values (as shown in (B)), resulting in near-optimal selection of TIs.

FIGURE S2. Numerical simulations comparing static and adaptive acquisitions for values of T_2 which extend outside

the assumed range of $T_2 \in [0.05, 0.45]$ Sec (shaded blue range). Comparisons for values of T_2 far below 0.05 s. could not be carried out because the static choice of echo times failed to fit. As T_2 increases beyond 450 ms, the adaptive acquisition continues to outperform the static one.

FIGURE S3. Histograms of optimal TEs for several different values of T_2 in the range [50, 450] ms. The dashed yellow line corresponds to $\text{TE} = \text{TE}_{\min} = 40$ ms, while the dashed red line corresponds to $1.4 \times T_2$. The optimal TEs are placed to an excellent approximation at $\text{TE}_{\min}$ and $1.4 \times T_2$ at a 1:5 ratio. Some deviation is observed for very short T_2 values, close to $\text{TE}_{\min}$.

FIGURE S4. The dynamics of TE and T_2 are shown for five simulations with $T_2 = 200$ ms, $\text{SNR} = 20$. Each simulation is color coded with a distinct color, and differs by the randomly generated Gaussian noise at each time step.

FIGURE S5. Simulation showing the dynamics of T_2 and the prior $p(\text{PD}, T_2)$ for a sample adaptive acquisition (true $T_2 = 200$ ms, $\text{SNR} = 20$). Top: The mean (solid, purple) and standard deviation (shaded purple) of $s(\text{PD}, T_2)$ with respect to T_2 at each excitation step. Bottom: the prior $s(\text{PD}, T_2)$ plotted at four excitations (indicated by black arrows on top). Above each 2D plot, its projection on the T_2 axis is shown; the center of this distribution corresponds to the mean T_2 , and its width corresponds to the shaded region (T_2 SD) for the top plot. This plot shows how the prior converges in real time very early on to approximately the correct solution. It also demonstrates how the width of the prior can be used to assess the precision of the estimate in real-time, and potentially be used as a stopping criterion for adaptive acquisitions.

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